RETAILVISION AI

Turning supermarket cameras, POS and inventory data into a retail decision engine

SEE	CONNECT	ACT
Customer & shelf behavior	POS + inventory context	Recommendations & alerts

MANAGER OBJECTIVE

Evaluate whether RetailVision AI is strong enough to justify a controlled pilot, and what evidence we need before committing meaningful engineering or commercial resources.

Prepared as a startup concept and business case. Financial figures and adoption assumptions are explicitly labeled where they are illustrative rather than validated.

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01 | EXECUTIVE DECISION MEMO

The opportunity in one page

A decision-oriented framing for management review

CORE THESIS

Supermarket systems tell retailers what was sold. RetailVision AI adds the missing context: what shoppers saw, touched, considered, returned, where shelves failed, and what operational action should happen next.

1	3	4	1
operating system	data streams	launch phases	pilot first

Problem: Store operators have fragmented visibility across CCTV, POS, inventory and staffing. The result is delayed decisions around stockouts, layout, conversion, labor and merchandising.

Solution: Use edge computer vision on existing camera infrastructure, correlate anonymous interaction events with POS and inventory data, and convert those signals into simple actions for store teams.

Business model: Start with a low-friction SaaS subscription priced by store/camera tier, then expand into integrations, predictive analytics, multi-store orchestration and supplier intelligence.

Recommended decision: Approve a small, measurable pilot rather than a full build. The pilot should prove three things: 1) interaction detection quality, 2) business impact on selected shelves, and 3) willingness to pay.

SUCCESS GATE

Do not optimize for “impressive AI demos.” Optimize for a quantified before/after business outcome with retailer-verified data.

02 | WHY NOW

Retail economics are making operational intelligence valuable

Evidence from current retail strategy research, paired with our product hypothesis

Current market context supports the direction: McKinsey reports that grocery remains a structurally low-margin business and that European grocers continue to rank cost and margin pressure highly in 2026. It also reports that 47% of surveyed grocery CEOs listed AI and automation among their top three priorities, while many still struggle to convert AI investment into measurable EBIT impact. [1]

That gap is strategically important. A product that turns existing store data into specific, operational actions can be easier to adopt than a broad “AI transformation” program. This aligns with the idea of starting with targeted, measurable use cases rather than attempting end-to-end transformation immediately. [1]

Competitive validation also exists: Trigo markets computer-vision retail intelligence based on anonymous shopper movement and product interactions, while Shopic positions computer vision across loss prevention, smart carts and in-store insights. [2][3]

OUR STRATEGIC ANGLE

We should not position RetailVision AI as “another autonomous checkout company.” The stronger wedge is affordable, retrofit-friendly retail intelligence for operators who already have cameras, POS and inventory systems.

47%	4-6 pts	60M+
CEO priority signal	AI EBITDA potential in McKinsey analysis of European grocery	annual activities cited by Trigo for model improvement

Note: These are external context signals, not guarantees of RetailVision AI outcomes. Our own market size, ROI and pricing assumptions remain hypotheses until validated.

03 | THE PROBLEM

The “before-purchase” blind spot: what stores cannot see today

POS tells the manager what was sold. Inventory tells what remains. Cameras show movement. The business problem is that these signals are disconnected, so managers often react after revenue has already been lost.

System	What it knows	Business gap
POS can see	What was finally purchased	Too late to explain why shoppers rejected or never reached the product.
Inventory can see	What is recorded as on-hand	Does not reliably explain shelf-level availability or shopper friction.
CCTV can see	Movement and activity	Raw video is not a business decision and is hard to scale manually.
Managers can see	Exceptions they happen to notice	Human observation is inconsistent, delayed and expensive.

The core problems RetailVision AI is designed to solve - and how it solves them:

- 1. Lost conversion: A product may be picked up, examined, and returned repeatedly without a sale. RetailVision AI detects pick/return behavior, compares it with POS sales, and highlights likely friction such as price, placement, or assortment.
- 2. Revenue lost to stockouts: A POS system can show a product sold, but not that the shelf became empty at 6:30 PM while demand was still high. Computer vision detects shelf gaps and inventory data helps estimate the operational priority.
- 3. Weak store traffic: Managers often cannot see which aisles customers bypass. Anonymous movement tracking creates aisle traffic and dwell-time patterns, allowing layout or merchandising experiments to be tested.
- 4. Missed cross-selling: POS can show items bought together, but behavior data can add context about the path to purchase. RetailVision AI can surface product relationships and placement opportunities such as “bread + butter” or “drinks + snacks.”
- 5. Delayed action: Dashboards often require managers to interpret multiple reports. RetailVision AI converts the signals into prioritized actions such as “restock Shelf 4 now,” “open another checkout,” or “test a placement change.”

Business problem	How RetailVision AI responds	Manager-level outcome
Stockouts on fast sellers	Detect shelf gaps, correlate with POS velocity and inventory, then trigger a restock priority.	Protect sales during the hours when demand is highest.
Customers show intent but do not buy	Detect pick/return patterns and compare them with transactions.	Identify products that need a pricing, placement, or assortment test.
Cold aisles and weak layouts	Map anonymous traffic, dwell time, and aisle flow.	Run evidence-based merchandising and layout changes.
Checkout queues	Monitor queue length and waiting-time signals.	Rebalance checkout staffing before queues become a customer-experience problem.

Slow-moving inventory	Combine dwell/interaction signals with inventory age and sales.	Identify candidates for promotions, relocation, or assortment review.
Data is fragmented across systems	Unify camera events, POS transactions and inventory records into one analytics layer.	Give the manager one operational view instead of several disconnected reports.

Core mechanism: camera events describe what shoppers do; POS confirms what was actually purchased; inventory describes what is available; the synthesis layer turns those facts into a ranked business action. The product is therefore not “a camera analytics tool” - it is a decision system for store operations.

STARTUP INSIGHT

The valuable asset is not “video analytics” by itself. It is the structured decision layer created by correlating behavior, commerce and availability.

04 | PRODUCT VISION

From camera footage to a store decision engine

Three inputs become one operational intelligence layer

CAMERA	POS	INVENTORY	AI SYNTHESIS	ACTION
Anonymous movement, dwell, pick/return, shelf state	Basket, transaction time, SKU sales	On-hand, replenishment, availability	Correlate + diagnose + prioritize	Alert, task, recommendation, report

The product should feel less like an analytics dashboard and more like an assistant for the store manager: “Here are the 3 things that need attention right now, why they matter, and the expected commercial effect.”

DESIGN PRINCIPLE

Every insight should answer: What happened? Why did it happen? What should the store do? How will we know it worked?

05 | PRODUCT USE CASES

Start with measurable pain, then expand

MVP to advanced intelligence

Use case	What the AI measures	Decision unlocked
A. Store flow	Footfall, peak hours, aisle heatmaps, dwell time	Manager sees where traffic concentrates and where shoppers disengage.
B. Shelf execution	Empty shelf/void detection, misplaced items, restock alerts	Staff gets an operational task before a stockout becomes lost revenue.
C. Pick-to-purchase	Pick, put-back and conversion signals	Merchandising sees high-intent/low-conversion products.
D. Inventory & demand	Forecast demand vs available stock	Store gets pre-emptive replenishment recommendations.
E. Layout optimization	Placement impact by zone and journey	Manager tests location changes using measured outcomes.
F. Queue operations	Queue length and estimated waiting time	Staffing is aligned to actual traffic patterns.
G. Daily AI brief	Three priority actions, exceptions, trends	Management receives a decision-ready summary without opening a dashboard.

MVP RULE

The first release should prove one workflow end-to-end rather than ship every feature. A strong candidate: shelf availability + traffic + POS correlation.

06 | CUSTOMER STRATEGY

Who actually pays?

Prioritize the buyer who owns the operating pain

Persona	Role	Value proposition
Primary buyer	Regional / independent supermarket owner or operations head	Needs: margin, stock availability, labor efficiency, simple reporting
Daily user	Store manager / shift manager	Needs: alerts, task prioritization, visual exceptions
Influencer	IT / POS / security lead	Needs: retrofit compatibility, integration, security, low operational burden
Expansion buyer	Multi-store COO / category leader	Needs: benchmarking, roll-ups, cross-store optimization
Future sponsor	FMCG brand / supplier	Needs: category visibility and measurable in-store execution

Ideal customer profile (initial): a store that already has functioning CCTV, basic POS data and inventory records, but lacks a unified analytics layer and has a manager who can act on daily recommendations.

AVOID

Starting with a customer who needs a full CCTV replacement, bespoke ERP rebuild, or heavy hardware installation. Those requirements destroy the “low-friction” wedge.

07 | COMPETITIVE STRATEGY

Where we can win

Compete on deployment economics, actionability and data integration

Category	Typical strength	RetailVision positioning
Enterprise autonomous retail	High automation / high complexity	Not the first battleground
Retail computer vision intelligence	Directly validated category	Differentiate with SMB affordability + retrofit
Generic CCTV analytics	Easy deployment	Weak business context without POS/inventory
BI / dashboards	Strong reporting	Weak real-time shelf/behavior signals
Retail ERP suites	Strong transactions	Often lack physical-world visibility

Competitive evidence: Trigo explicitly offers retail intelligence from shopper movement and product interaction, including pick/put-back and placement insights. Shopic combines computer vision with loss prevention, smart-cart and store-insight products. [2][3]

POSITIONING STATEMENT

RetailVision AI is the practical intelligence layer for supermarkets that already have cameras but need the business meaning behind the footage - without replacing the store’s core systems.

08 | DEFENSIBILITY

The moat is the learning loop, not the camera model

Build compounding advantages over time

DATA MOAT Store-specific event histories, product interactions, shelf states and outcomes create proprietary training/evaluation data.
WORKFLOW MOAT Once daily decisions, alerts and tasks are embedded in store routines, switching costs rise.
INTEGRATION MOAT Reliable POS/ERP/camera connectors become reusable deployment assets.
MODEL MOAT Store-aware calibration, shelf geometry, occlusion handling and event correlation improve with deployment learning.
BENCHMARK MOAT Across many stores, anonymized benchmarks can reveal expected patterns by format, time and category.
IMPORTANT BOUNDARY Do not claim a “data moat” before collecting differentiated data at scale. The moat is an intended strategy and must be earned through deployment.

09 | TECHNOLOGY ARCHITECTURE

Edge-first, cloud-connected, privacy-by-design

Keep real-time processing close to the store

Layer	Candidate technologies	Purpose
Edge CV	Python, OpenCV, YOLO-family detector, tracker, PyTorch/TensorRT	RTSP decoding, anonymous tracking, shelf events
Backend	FastAPI, Redis/Celery	Event ingestion, alerts, orchestration
Data	PostgreSQL + time-series/spatial extensions	Events, store topology, aggregates
Frontend	React / Next.js + charting	Dashboard, heatmaps, alert feed
Cloud	AWS / GCP / Azure	Multi-tenant SaaS, model registry, monitoring
Hardware	NVIDIA Jetson / Intel NUC class	Local inference and privacy controls

The original blueprint proposes Python/OpenCV/YOLOv11 + tracking, FastAPI/Redis, PostgreSQL/TimescaleDB/PostGIS, React/Next.js, and edge hardware such as NVIDIA Jetson with cloud synchronization. These are candidate technologies, not locked architectural decisions. [Source blueprint]

ARCHITECTURE PRINCIPLE

Store-local inference should minimize bandwidth, latency and privacy risk; cloud services should focus on aggregation, configuration, learning and multi-store management.

10 | TRUST, PRIVACY AND GOVERNANCE

Make “privacy by design” a selling point

Trust has to be part of product architecture and sales

No facial recognition: Do not identify individuals; use anonymous session IDs only.

Edge processing: Prefer local processing for raw video; avoid exporting unnecessary footage.

Minimal retention: Store event data needed for analytics; define explicit retention policies for any diagnostic media.

Role-based access: Separate store, region and corporate permissions.

Auditability: Log model decisions, alerts and user actions so recommendations can be reviewed.

Legal review: Adapt controls to the laws and contracts applicable in each deployment region; do not treat “GDPR-ready” as a blanket certification.

COMPETITIVE TRUST SIGNAL

Trigo also emphasizes anonymous tracking and privacy-oriented retail computer vision. Privacy therefore should be treated as a category expectation, not a differentiator we can assume without execution.

11 | MVP SCOPE

The smallest product that can prove the business

Cut everything that does not help the pilot produce a measurable outcome

Bucket	Scope	Reason
Must have	Camera ingestion; customer counting; aisle traffic; shelf void detection; basic POS correlation; alerts; simple dashboard; daily summary	Prove the core workflow end-to-end.
Nice later	Pick/return vectors; demand forecasting; price recommendations; queue balancing; shrink review	Add only after the core value loop is validated.
Do not build first	Full autonomous checkout; custom hardware; broad ERP marketplace; consumer app; complex personalization	Avoid scope that increases capital or integration burden before proof.

PILOT HYPOTHESIS

If a store receives earlier stockout alerts and clearer aisle/traffic insights, it should improve on-shelf availability or operational response without requiring new POS hardware.

12 | PILOT DESIGN

Turn the idea into evidence in 6-8 weeks

A pilot should be a business experiment, not a product launch

Time	Stage	Output
Week 0-1	**Baseline**	Select store, map cameras, define 10-20 SKUs/zones, capture baseline stockout/traffic metrics.
Week 2-3	**Instrument**	Deploy edge processing, validate events against manual samples, connect POS/inventory data.
Week 4-6	**Operate**	Run alerts and recommendations; store managers act and record outcomes.
Week 7-8	**Evaluate**	Compare before/after performance, document false positives, quantify value, interview users.

Pilot KPIs: shelf-availability rate, time-to-restock, alert precision, store-manager time saved, conversion proxy for selected products, daily active usage, and willingness to pay.

GO / NO-GO LOGIC

Go: measurable operational improvement + credible willingness to pay + repeatable deployment. No-go or pivot: weak data quality, excessive false alerts, or value that cannot be monetized.

13 | BUSINESS MODEL

Monetize the insight layer, not the hardware

Simple entry pricing with expansion paths

Illustrative price	Tier	Included value	Role in model
$149 / store / month	Starter	Traffic, counting, heatmaps, weekly report	Acquisition / proof
$349 / store / month	Pro	POS + inventory integration, shelf alerts, conversion funnels, daily AI digest	Core commercial offer
$699+ / store / month	Enterprise	Multi-store controls, advanced predictions, custom integration	Expansion

These prices come from the original concept blueprint and should be treated as starting hypotheses to test, not market-validated rates. [Source blueprint]

EXPANSION LADDER

Store -> multi-store -> advanced forecasting -> workflow automation -> supplier/category analytics. This increases ARPU without forcing the first sale to carry the entire roadmap.

14 | FINANCIAL LOGIC

A disciplined way to think about SaaS economics

Use assumptions, not hype

$349	70%	3-6 mo
illustrative Pro price	illustrative gross margin target*	pilot-to-payback aspiration*

*Internal planning targets, not validated outcomes.

Metric	Formula	Illustration / management question
Monthly recurring revenue	Stores x ARPU	Example at 100 Pro stores: 100 x $349 = $34,900 MRR
Acquisition cost	Sales + install + onboarding / new store	Track by channel and separate one-time deployment cost
Gross margin	Revenue - cloud/edge/service cost	Target higher margin as model inference and support become standardized
Net retention	Expansion - churn	Key signal that integrations and advanced modules are valuable
Payback	CAC / monthly gross profit per store	Business should converge toward a predictable payback window

MANAGER TAKEAWAY

The business becomes attractive when deployment is repeatable, support load stays low, and each additional store adds more recurring revenue than variable operating cost.

15 | ROI STORY

Sell the outcome, then prove the number

A sample ROI framework for a pilot

The original blueprint uses an illustrative supermarket with $150,000 monthly revenue and estimates $4,500/month stockout recovery plus $2,200/month from layout/price optimization, versus a $349 Pro subscription. That yields a stated gain-to-subscription ratio above 19x. [Source blueprint]

For management review, that figure should be treated as a scenario, not a promise. The pilot must replace it with observed values using the store's own data and attribution rules.

$6,700	$349	>19x
illustrative monthly gain	illustrative Pro fee	illustrative gain / fee

Recommended ROI measurement: define a control period, select a narrow set of SKUs/zones, document interventions, and separate changes plausibly caused by the product from normal seasonality, promotions and supply changes.

WHAT MAKES ROI CREDIBLE

A store owner does not need a perfect AI accuracy score. They need a believable chain from signal -> action -> measurable operational or commercial result.

16 | GO-TO-MARKET

Sell through proof, not a giant sales deck

Land with one store, expand through measurable value

Motion	Target	How
1. Design partners	5-10 stores	Discounted pilot or low-cost proof in exchange for data access, operator feedback and references.
2. Local channel partners	CCTV installers, POS providers, retail consultants	Use trusted deployment relationships to reduce integration friction.
3. Case-study engine	Before/after evidence	Turn successful pilots into quantified sales stories and manager testimonials.
4. Multi-store expansion	Regional chains	Standardize connectors and onboarding, then expand account-wide.
5. Supplier analytics	Brands / distributors	Offer aggregated category insights only after privacy, permissions and data governance are mature.

BEST FIRST SALES ASSET

A one-page pilot scorecard showing baseline vs post-deployment outcomes, not a list of AI features.

17 | PRODUCT ROADMAP

Four phases, with evidence gates

Build in sequence so each phase funds the next

Phase	Theme	Deliverables	Evidence gate
Phase 1 | Months 1-3	Flow & visitor intelligence	Footfall, tracking, aisle heatmaps, peak hours	Gate: stable event quality in real store conditions
Phase 2 | Months 4-6	Shelf watch	Shelf presence, void detection, restock alerts	Gate: actionability and low false-alert burden
Phase 3 | Months 7-9	POS triangulation	Basket correlation, pick-vs-sale funnels, recommendations	Gate: measurable commercial or operational lift
Phase 4 | Months 10-12	Predictive optimization	Demand forecast, inventory prediction, dynamic recommendations	Gate: forecast accuracy + willingness to pay for advanced tier

PRODUCT PRINCIPLE

Do not move to the next phase because the calendar says so. Move because the previous phase has a validated quality and business-value gate.

18 | OPERATING DASHBOARD

What management should watch every month

A small KPI system keeps the startup honest

KPI area	Metrics	Question
Product quality	Detection precision/recall, false alerts, uptime, latency	Are we trustworthy in real stores?
Customer value	Stockout reduction, restock response time, conversion lift proxy	Are we improving store economics?
Adoption	Weekly active managers, alert acknowledgement, recommendation action rate	Do people actually use it?
Commercial	Pilot conversion, ARPU, churn, NRR, CAC payback	Can this become a scalable business?
Deployment	Days to install, integration hours/store, support tickets	Can we scale without services eating margin?

NORTH-STAR METRIC

Value realized per store per month: a composite view of measurable benefit relative to subscription and operational effort.

19 | RISKS AND MITIGATION

What can break the startup?

Address the failure modes before scaling

Risk	Why it matters	Mitigation
Vision accuracy in messy stores	Occlusion, reflections, camera angle, product similarity	Constrain initial SKUs/zones; calibration tools; human-verified samples.
Integration complexity	POS/ERP variability	Build connector abstraction; standardize a small number of platforms first.
False alerts	Alert fatigue kills trust	Confidence thresholds, prioritization, feedback loop, rate limits.
Privacy / legal	Regional surveillance and data rules	Privacy-by-design, local counsel, explicit retention/access controls.
Hardware/support burden	Every store behaves differently	Favor existing CCTV; standardize supported edge devices; remote health monitoring.
Weak attribution	Sales move for many reasons	Use controlled pilots, test zones and intervention logs.
Long enterprise sales cycles	Small team cannot carry heavy sales process	Start with regional operators and partners; land small, expand after evidence.

CRITICAL LESSON

The highest risk is not “can we train a model?” It is whether the product can produce reliable signals, integrate with existing operations, and cause a manager to take a profitable action.

20 | TEAM AND CAPABILITY PLAN

What the startup needs to execute

Keep the initial team small but cross-functional

Capability	Primary accountability
Founder / product lead	Customer discovery, roadmap, pilot ownership, pricing
ML/CV engineer	Detection, tracking, event models, evaluation
Backend/data engineer	Integrations, event pipeline, analytics, APIs
Frontend/product engineer	Manager dashboard, workflows, alert UX
Retail advisor / part-time	Store operations, merchandising, validation
Security/privacy advisor / part-time	Deployment controls, contracts, governance

BUILD VS BUY

Buy commodity infrastructure where possible. Build the differentiated event correlation, retail workflows, store-aware models and recommendation layer.

21 | THE ASK

What I would ask management to approve

A concrete path from idea to evidence

DECISION REQUESTED

Approve a controlled pilot with 1-3 stores, a defined evaluation window, and access to the minimum camera/POS/inventory data required to measure business impact.

Pilot budget should be capped and tied to milestone gates rather than approved as an open-ended product build. The management sponsor should nominate a store operator as the single pilot owner and agree on the baseline metrics before deployment.

[Commercial] Identify 1-3 design-partner stores and a buyer/user sponsor.

[Technical] Confirm supported camera feeds, POS export/API, and inventory data availability.

[Measurement] Agree on 3-5 KPIs and a baseline period.

[Governance] Approve privacy/security review and data-handling rules.

[Exit] Set go/no-go criteria for broader investment after the pilot.

EXPECTED OUTPUT FROM PILOT

A quantified case study, validated technical architecture, prioritized roadmap, realistic price point, and a clear recommendation to scale, pivot, or stop.

22 | SOURCES AND ASSUMPTIONS

What is established vs what must be validated

Transparency for manager and investor conversations

[1] McKinsey & Company, “The State of Grocery Retail Europe 2026” and related 2026 retail AI research. Used for context on grocery margin pressure, AI priorities and potential value pools.

[2] Trigo Retail, company pages on retail intelligence and technology. Used to validate that computer-vision shopper/product interaction and anonymous tracking are established solution patterns in the category.

[3] Shopic, company site. Used to validate current category activity around computer-vision retail intelligence, loss prevention and store insights.

[4] Source blueprint supplied by the user: “RetailVision AI - Global Master Blueprint.” Used for the original product concept, technical stack candidates, phased roadmap, pricing examples and illustrative ROI scenario.

ASSUMPTIONS THAT MUST BE VALIDATED

Market size; SMB share; willingness to pay; actual ROI; model accuracy across store formats; installation economics; integration effort; legal requirements by country; churn and expansion behavior.

MANAGER-FRIENDLY RULE

Any number that is not backed by a cited source or a real pilot should be labeled as an assumption, scenario or target.

End of manager review draft.